Zhiao (Zachariah) Jia

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EDUCATION

Calvin University Grand Rapids, MI
Bachelor of Science in Engineering, Electrical and Computer Engineering *August 2022 — May 2026*
- GPA: **3.73**/4.0 | Minors: Computer Science & Mathematics | Dean's List (6 semesters)

PROJECTS

Laser-Based Goose Repeller Fall 2025 — Spring 2026
- Led a 4-engineer team to integrate an autonomous perception-to-actuation system on a Raspberry Pi embedded controller, unifying vision, motion control, mechanical, and power subsystems into a live robot.
- Fine-tuned a computer-vision model (Ultralytics YOLO26 nano) on a self-curated 4-class dataset (**~2,000** images), reaching **0.85** precision; ensured safety by adding non-target classes (human, cat, dog) to suppress false actuation.
- Programmed a vision-guided 2-DOF pan-tilt laser controller in Python via hardware PWM to steer toward detected goose, with a randomized offset to counter habituation.

Multi-cycle Processor Design Spring 2024
- Designed and implemented a multi-cycle NIOS II subset processor (**20** instructions) in VHDL on Altera DE2 (Cyclone II) board.
- Reduced effective CPI from **5 to 2** (**60%** reduction), compared to course reference, for R/I-type and memory-related instructions by re-designing the control path as a 3-state FSM.
- Achieved **45.28MHz** maximum frequency (Fmax) with 1059 registers; integrated DE2 peripherals (switch/button input, 7-segment/LED output) for end-to-end I/O.

CAN-Based Angle Error Detection System Spring 2024
- Developed a dual-Arduino servo fault-detection system (Arduino C/C++, 500kbps CAN bus) comparing commanded angles against MPU6050 IMU feedback to flag servo faults, validated by injecting failures and confirming detection.

Distributed MQTT Smart Home Monitoring System Fall 2024
- Engineered and built a bidirectional MQTT pub/sub system over 3 microcontroller nodes (Raspberry Pi Pico, Pi Zero) and a Mosquitto broker, with a request/response layer letting a button-driven LCD node query sensor and camera data on demand.

WORK EXPERIENCE

Engineering Summer Research Assistant Summer 2024 — Summer 2026
Calvin School of STEM *Grand Rapids, MI*
- Built an automated geospatial damage-assessment pipeline, published as use-case documentation on DesignSafe (NSF natural-hazards cyberinfrastructure), processing over **70,000** disaster images through detection, GPS extraction, NWS weather enrichment, and interactive mapping.
- Led a multi-summer computer-vision research project on flipped-vehicle detection; curated a labeled dataset (**~1,500** images) and trained a YOLO computer vision model for automated disaster-imagery damage assessment.

Engineering Lab Teaching Assistant September 2023 — May 2026
Calvin Engineering Department *Grand Rapids, MI*
- Evaluated student work and provided feedback on RISC-V ISA, 5-stage pipelined CPU design (hazards, forwarding, branch prediction), and memory hierarchy (cache policies, virtual memory, TLB).
- Delivered **280+** hours of lab instruction across 6 semesters, mentoring 20+ students per session in three courses covering VHDL/FPGA design, multi-cycle CPU architecture, NIOS II assembly, state machines, and analog/digital circuit labs.

IT Helpdesk Technician October 2023 — March 2026
Calvin Information Technology *Grand Rapids, MI*
- Diagnosed component-level hardware faults (RAM/power failures) via symptom analysis against technical documentation.
- Imaged and deployed campus desktops using SCCM; resolved on-site classroom AV and system failures during active sessions.

SKILLS

- **Lab**: Digital Multimeter, Oscilloscope, Function Generator, Power Supply, Soldering Iron.
- **Programming**: C, C++, VHDL, NIOS II Assembly, Python, Bash
- **Tools**: Quartus, ModelSim, Altera Monitor Program, PSpice, Eagle CAD, MATLAB, Git, Linux, Wireshark

CERTIFICATIONS

Getting Started with AI on Jetson Nano, NVIDIA April 2026
PCB Basic Design Course, Altium Education April 2026